

The poor and limited coverage even in metros is because the country still doesn't have adequate number of 3G-enabled towers. According to a report, of the 350,000 towers, only 60,000 are 3G-enabled, which means that the promised speed is not available to everyone. The inconsistency in the speed as well as quality can be attributed to lack of a fibre-optic system which would have supported high bandwidth of the 3G technology—a must for high-bandwidth networks.

To deploy 3G and 4G, telcos have been using microwave links to support the towers. And while 3G network was supposed to offer data speed of 3Mbps to at least 21Mbps, the actual speed was lower than 1Mbps, blurring the line between 2G and 3G. At best, 3G was used to stream audio and videos at an agonisingly low speed.

This explains why 4G failed to take off in India. The jump from 2G to 4G without working on the requisite infrastructure, i.e., fibre-optics base for 3G, did hamper 4G with quality of service and data speed remaining disappointingly low.

However, network providers have now realised that in order to successfully implement the next generation of wireless communication, i.e., 5G, they would need to address this gap. And thus, they have started investing in fibre-optic communications extensively. As per industry experts, currently, India is installing only 15 million kilometres of fibre every year, as against 50 million kilometres of fibre required for efficient and effective implementation of both 3G and 4G services.

Reliance Infotel is setting up its own fibre-optic base for not only inter-city communication but intra-city services as well. Nevertheless, installing the fibre-optic base is not an easy process as it

means digging at several places to put fibre ducts across the city. This could be one plausible reason why telcos did not switch to fibre-optics backbone earlier.

5G: Taking it to the next level

The next-generation wireless broadband technology will make the world smarter. From multimedia messaging to video streaming, from connected vehicles to autonomous driving, from Nano IoT to implantable wearable—every application is expected to move to the next level.

5G is hopping to a new level where innumerable devices can be connected to each other using D2D standard and where security breach is extremely difficult. This D2D technology would be beneficial even when the network is out of range. A wide array of sensors network can be used for home security systems, energy grids and entertainment.

Basically, 5G is 4G technology integrated with Worldwide Wireless Web. Currently, in terms of bandwidth, it offers more than 1Gbps as against 200Mbps offered by 4G.

Communications sector is going to see a massive increase in demand once 5G takes shape. Therefore to supersede the existing technology, 5G needs to lower its bit rate value. While we have discussed data speeds of both 3G and 4G earlier in the article, 5G is expected to be at least ten times faster than 4G, i.e., at least 10Gbps. In order to support such high bit rate, the latency (the time taken to stream a video or any other function) needs to be very low at around one millisecond.

However, a few telco players have ambitious expectations as far as 5G is concerned. According to a report by 5G.co.uk, experts project that 5G networks will be capable of data transfer at speeds of up to

20Gbps. Similarly, Nokia expects 5G to enable streaming of an 8k video in 3D.

Experts suggest that 5G will provide ultra-fast, ultra-reliable and ultra-high-capacity transmission at low latency, making communication and streaming faster.

The current scenario

Globally, work has started on 5G. The US is one of the first countries to usher in 5G by opening up the high-band spectrum for the technology. According to reports, the Federation Communication Commission (FCC) in the US has already started work in the area.

Samsung Electronics America along with Cisco partnered with American telecom major Verizon to announce 5G trials in eleven cities of the US back in February this year. The multi-vendor end-to-end 5G trial was the first in the field that was launched in Michigan. Trials for at least five cities in the US are scheduled by the second quarter of FY18, while a total of eleven pilot trials will be carried out by the end of 2017.

The trial network has a 5G virtualised packet core as part of the Cisco Ultra Services Platform with Cisco Advanced Services and Samsung virtual RAN (vRAN) solutions. It has been paired with Samsung's 5G radio base stations and 5G home routers, which will deliver broadband services to Verizon's customers who have signed up for the trials.

The three companies have followed Verizon's 5G Technical Forum specifications—a series of network vendor interoperability tests to demonstrate seamless inter-working between core network, radio edge and user devices to exhibit next-generation network virtualisation via multi-vendor support.

"We're very excited about the possibilities this opens up in